CornerBrief PRD

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CornerBrief turns a NYC street corner into a review-ready brief showing what is known, what is missing and what an engineer should check on site. It assists an engineer and does not approve designs or claim compliance.

Problem and goal

The demo must prove that a multi-agent system can support early engineering investigation without hiding uncertainty. Established tools already help design and inspect ramps, so CornerBrief focuses on the research that comes before detailed design.

- **Job it supports:** The Junior Engineer (Design) posting includes research studies, engineering investigations, field surveys and design work under an Engineer-In-Charge.
- **Why corners:** DDC runs a pedestrian ramp unit, and some ramps need custom designs because of obstacles such as private utilities.
- **Differentiator:** The tool labels every input as measured, assumed or missing, and it stops when the data is too thin.

The team has adopted the street corner direction, with one added requirement: a strong rule set covering both the NYC and New York State codes. "CornerBrief" is a working name.

Users and scope

The primary user is a junior engineer preparing a corner for review, and the engineer in charge reviews the result.

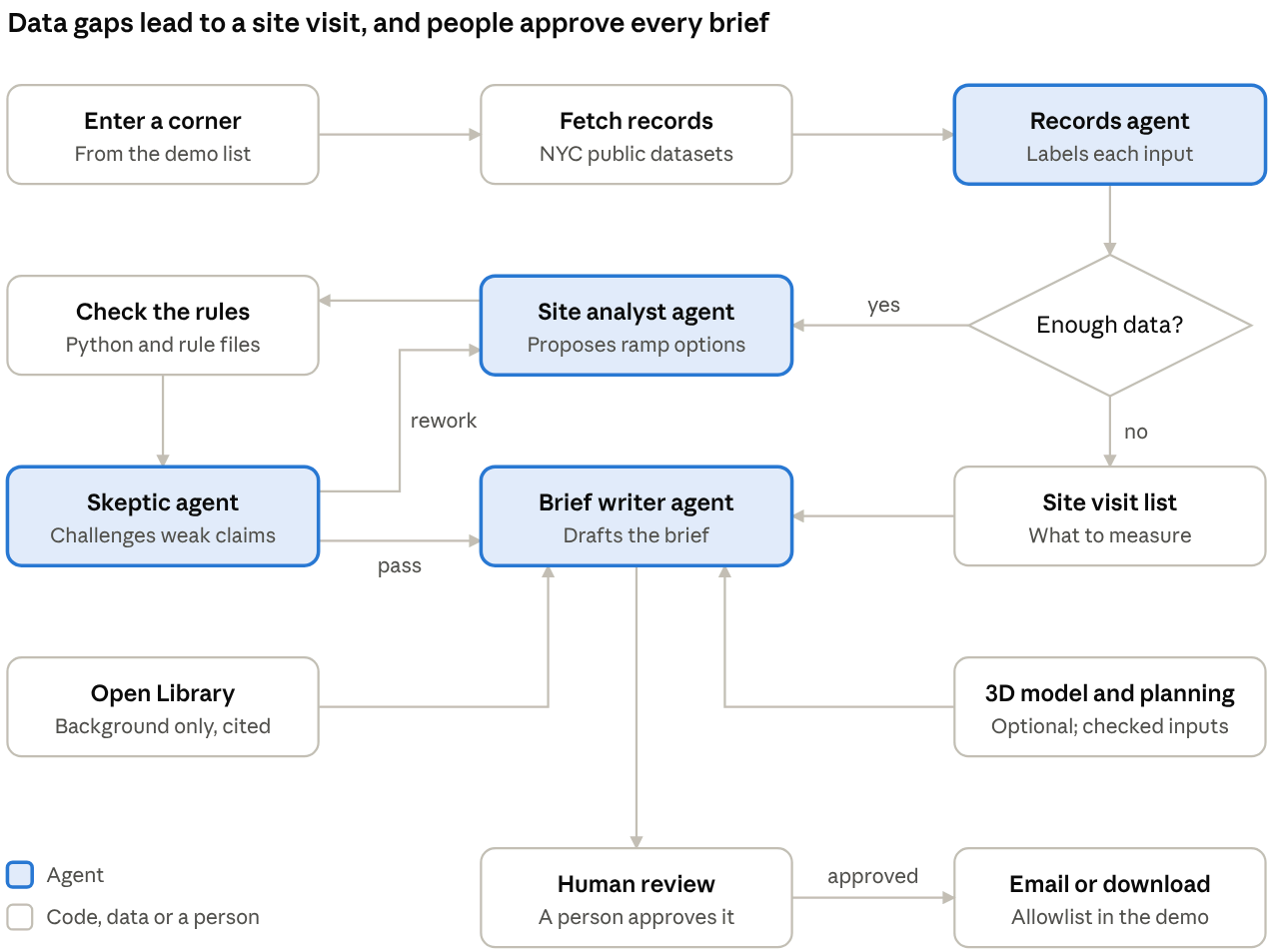
In scope for the demo	Out of scope
Three real NYC corners, with raw records saved locally	Citywide batch processing
Standard, custom or unknown sorting	Final design or construction drawings

In scope for the demo	Out of scope
Rule set with one file per rule and an exact source section	Declaring a corner compliant, approved or safe
An applicable-law decision for every corner	Legal advice or replacing an engineer's judgment
Stale-data detection using the survey and construction dates	Replacing surveys or field inspection
Site visit list	Filing plans or technical infeasibility forms
Simple browser 3D model and takeoff of modeled parts	Detailed CAD, photorealistic rendering or structural design
Concept cost range, labor hours, equipment types and work sequence for one corner	Bids, contract schedules, crane selection or crane positions
Safety questions for a professional	Site safety plans or final PPE lists
Background reading from Open Library	Using books as rule or design inputs, or processing borrowed in-copyright text
Download, and approved email to test addresses	Reading email replies or sending to real engineers

Future ideas, not part of the demo: urban gardening site briefs, an architectural concept visualizer, and a separate book-grounded design and estimate tool for larger projects such as the LaGuardia garage concept.

System diagram

The flow splits as soon as the records are labeled, so a corner with too little data never receives a design suggestion.



CornerBrief system flow · 4 agents, 1 data check, 1 approval step

A failed rule check or weak claim returns to the site analyst, with a retry limit. Python decides which rules apply for every corner, including corners that go straight to a site visit, and that decision appears in the brief. Open Library and the 3D model feed only the brief, and neither can change a rule result.

Components and modules

The initial prototype builds the core modules first, then adds planning for one corner if the core tests pass. Plain code does every calculation, and all planning results come from one measured takeoff.

Module	What it does	Run by	Prototype stage
Data loader	Loads saved ramp, progress and complaint records	Code	Core
Evidence checker	Handles placeholders, units, stale data and evidence labels	Code and Records agent	Core
Rule library	Stores one file per rule, plus precedence files	Code	Core
Applicability resolver	Decides which rules apply and records why	Code	Core
Site analyst	Proposes likely constraints and ramp options	Agent	Core
Rule checker	Runs limits and returns the four allowed results	Code	Core
Skeptic	Challenges weak sources, conflicts and claims	Agent	Core
Site visit planner	Lists missing inputs and what to measure	Code	Core
Brief writer	Drafts the brief from validated results	Agent	Core
Review and export	Requires approval and offers a download	Code and person	Core
Parts kit and 3D model	Builds simple ramp, landing and curb geometry	Code	Add-on 1
Takeoff	Measures modeled area, volume and length	Code	Add-on 1
Cost	Applies dated rates and shows a concept range	Code	Add-on 2
Crew and sequence	Estimates labor hours by trade and orders the work	Code	Add-on 2

Module	What it does	Run by	Prototype stage
Equipment types	Lists likely equipment types, with no crane choice or position	Code	Add-on 2
Safety questions	Lists silica, traffic and training items for a professional to check	Code	Add-on 2
Library	Adds cited public-domain background reading	Code and Brief writer	Add-on 3
Email	Sends approved briefs to allowlisted test addresses	Code and person	Add-on 3

Cost rates and crew rates live in separate dated files, each marked as sample, public or licensed. Planning results use the label "Class 5 concept estimate, not a bid" and list what the model excludes.

Compliance rule set

The rule set covers NYC and New York State codes, but it must first decide what applies. For a street corner in NYC, the state Uniform Code does not apply, and the NYC Construction Codes mainly govern buildings. The limits checked at the corner come mostly from NYC street rules and the ADA.

Rule file requirements

- **One rule per file:** Each file lists its limit, unit, jurisdiction, authority type, edition, effective date and scope.
- **Exact source:** Each file gives the section, a link, a short quote and the date it was retrieved.
- **No recited law:** Models cannot create, edit or supply limits, and a person reviews each rule.
- **Status controls checks:** Only `verified` rules can produce a limit result. A rule marked `needs_source` returns "unknown."
- **Careful result wording:** Results are "limit exceeded," "no conflict found," "unknown" or "not applicable."

Which rule applies

These decisions live in reviewed precedence files, each with its own source and reasoning. The brief shows the rule chosen and the reason.

Question	Decision	Why
NYC or NYS building code at an NYC location?	NYC; the Uniform Code does not apply	NYS says the Uniform Code excludes NYC, which has its own construction code
Same question outside NYC?	NYS Uniform Code	It applies in every part of the state except NYC; check for approved local additions
Corner ramp in the public right-of-way?	ADA, NYC Administrative Code, Highway Rules and H-1011	The city's design guide and §2-09(f)(4)(xiv) point to these
Route from the sidewalk to a building?	NYC Building Code, as a flag for engineer review	§1104.1 covers routes within a site, not the ramp itself
A rule tier is stricter than another?	Flag both, and use the stricter for the warning	The tool highlights the conflict, and an engineer decides
State-controlled road in NYC?	Unknown until ownership is confirmed	NYSDOT's manual may matter on state projects, so the tool should not assume

Starter rules

These sections were checked against the linked text. A person should still confirm each one before use.

Rule	Limit	Exact source	Status
Curb ramp slope	8% maximum	NYC Admin. Code §19-112	Verified
Curb ramp width	4 feet minimum	NYC Admin. Code §19-112	Verified

Rule	Limit	Exact source	Status
Lip at the roadway	5/8 inch maximum where a common level is not possible	NYC Admin. Code §19-112	Verified
Sidewalk transverse slope	1 to 3 inches in 5 feet (about 1.67% to 5%)	34 RCNY §2-09(f)(4)(xi)	Verified
Corner ramp standard	Must follow the latest H-1011	34 RCNY §2-09(f)(4)(xiv)	Verified
Recommended sidewalk slope	0.50% minimum; 2.1% maximum for at least 5 feet of width, 5% for the rest	NYC DOT filing guidelines, VIII.F.2	Verified; guidance
Gutter slope	0.50% minimum	NYC DOT filing guidelines, X.F	Verified; guidance
Ramp design margin	At least 0.5% below H-1011's acceptance slopes	NYC Street Design Manual	Verified; needs H-1011 values
H-1011 slopes and landing	Not yet available	Current H-1011	Needs source
Flare slope	9.5% maximum next to a pedestrian path	2021 H-1011 presentation	Needs current H-1011
ADA curb ramp limits	Not yet available	2010 ADA Standards, exact sections	Needs source
2025 NYS Building and Residential Code limits	Not yet available	Exact sections of the 2025 codes	Needs source; applies outside NYC

PROWAG is not a starter limit. USDOT adopted it for transit stops, and a February 2026 congressional letter says DOJ had not adopted it. Treat it as reference material unless an engineer or agency says otherwise.

Requirements

The demo is complete when every P0 requirement passes. P1 items should follow if time and budget allow.

ID	Requirement	Priority
R1	Accept a corner from the demo list and load its saved records	P0
R2	Label every input as measured, assumed or missing, with its source and survey date	P0
R3	Stop design suggestions and create a site visit list when required data is missing or stale	P0
R4	Suggest likely ramp options only when required data is present	P0
R5	Check values in Python against one file per rule, each with an exact source section	P0
R6	Have the skeptic challenge weak sources, conflicts and unsupported claims, with a retry limit	P0
R7	Produce a brief with findings, reasoning, unknowns, site checks and sources	P0
R8	Require a person to approve a brief before it can be emailed	P0
R9	Show that the output is pre-design support and not a compliance decision	P0
R10	Offer a brief download	P0
R11	Show a simple 3D model built from checked dimensions	P1
R12	Email approved briefs from the app's dedicated address to allowlisted test addresses only	P1
R13	Add cited Open Library reading in a section labeled background only	P1

ID	Requirement	Priority
R14	Record each agent's input, output and token use for debugging and cost tracking	P1
R15	Decide in Python which NYC, state and federal rules apply, then state the reason and source in the brief	P0
R16	Reject a rule file with no exact source, and return "unknown" for any unverified rule	P0
R17	Show both rules when tiers differ, use the stricter for the warning, and leave the decision to an engineer	P0
R18	Prevent models from creating, changing or supplying rule limits	P0
R19	Treat placeholder values such as `999` as missing, and require documented units before a limit check	P0
R20	Compare survey and construction dates, and mark measurements stale if a corner was rebuilt later	P0
R21	Measure modeled area, volume, length and counts from one takeoff	P1
R22	Calculate cost and labor hours in code from dated rate files with a source and status	P1
R23	Show likely trades, equipment types and work sequence, with no crane choice or position	P1
R24	List safety questions for a professional, without presenting them as a safety plan or final PPE list	P1
R25	Label planning results "Class 5 concept estimate, not a bid," show exclusions, and skip them when dimensions are not checked	P1

Agents and data

Four agents handle interpretation and writing, while regular code handles data fetching, math, rule checks and sending.

Agent	Job	Must not
Records	Organize corner records and label the evidence	Fill gaps with guesses
Site analyst	Identify likely constraints and ramp options	Supply a limit or decide whether a rule passes
Skeptic	Challenge weak sources, conflicts and unsupported claims	Override a Python check or applicability decision
Brief writer	Draft the brief for the engineer	Add facts, limits or legal conclusions missing from the record

Source	Rows	Last updated	Use in the demo	Limits
Pedestrian Ramp Locations (`ufzp-rrqu`)	217,679	Oct 2021	Ramp slopes, widths, curb lip, ponding, obstacles and survey date	Old; units are not documented; `999` appears as a value
Pedestrian Ramp Program Progress (`e7gc-ub6z`)	187,749	Nov 2026	Corner status, construction type and completion date	No measurements
Pedestrian Ramp Complaints (`jagj-gttd`)	6,571	Nov 2026	Complaints and temporary repairs	No coordinates; test the join on corner ID
Rule files	12 starter rules	Varies	Applicable limits and source sections	Some still need sources
Open Library	Not applicable	Not applicable	Optional background reading	Cite results; never use them as rule inputs

These counts come from the first data report, which read column details and the first 1,000 rows of each dataset. The full datasets still need to be checked. The report found no sidewalk width, legal grade or ramp type field, so those values must be marked missing or gathered from another source.

The app's dedicated email should use a sending service, with its API key stored in an environment variable. The Open Library user agent should include a shared contact address.

Demo and acceptance

The demo passes when all three test corners behave as expected, ending with the corner where the tool stops.

Test corner	Expected result	Pass criteria
Standard	Suggests a likely ramp type and shows the math	Hand-checked calculations match; every limit shows its exact source and why it applies
Constrained	Flags the corner as probably custom	Uses evidence such as a complex construction status or recorded obstacle, or marks a site check as needed
Missing or stale data	Refuses to propose a design	Lists the missing inputs or stale measurements, plus what to measure on site

These rules apply to every run:

- No brief claims that a corner is compliant or safe.
- No email leaves without a person's approval, and every recipient must be on the allowlist.
- A civil engineer reviews the rule file and the three briefs before the final demo.

The applicability tests must also pass:

- An NYC location rejects the NYS Uniform Code and cites the reason.
- A non-NYC test location selects the NYS rule files and uses "unknown" until their sources are verified.
- A corner ramp does not use a building code limit.
- A rule file missing its exact source section fails to load.

If the planning add-ons are built, they must also pass these tests:

- Hand-checked takeoff, cost and labor calculations match on one corner.
- All planning results use the same takeoff, and a dimension change updates all of them.
- A sample rate is labeled as sample wherever it affects a result.
- A missing rate returns "unknown" instead of a number.
- The missing or stale corner produces no takeoff or estimate.

Risks and open questions

The largest risks are outdated measurements and missing H-1011 values, because both can make a confident result wrong.

Risk	Response
Measurements predate a rebuild	Compare survey and construction dates, and mark them stale
Field meanings or units are misread	Confirm the data dictionary; use "unknown" until units are documented
Sidewalk width, legal grade or ramp type is unavailable	Mark it missing and add it to the site visit list
The wrong code is applied	Use reviewed precedence files, tests and engineer review
A rule is outdated or misquoted	Store edition, date, link and quote; expire rules that need review
H-1011 or NYS code text is unavailable	Keep those rules as `needs_source`; do not fill them from memory
An agent sounds more certain than the evidence	Keep evidence labels, the skeptic and Python checks in the flow
API costs run too high	Cache responses, use fixed corners and limit retries
Book passages get used as rules	Keep Open Library output in a separate section that cannot reach the checks
Email goes to the wrong person	Require approval and use an allowlist
An output misleads readers	Label it as a demo, make no compliance claim, and don't imply any NYC affiliation
An estimate is taken as a bid	Show the class label, rate sources, exclusions and sample-data warnings
Modeled quantities look complete	List excluded work such as utilities, drainage, traffic control and any unmodeled parts

Risk	Response
Planning output is taken as safety advice	Use questions for a professional, with no site safety plan, final PPE list or crane choice
Add-ons delay the demo	Build the core first, and drop add-ons before cutting core tests

Open questions

- [x] ~~Does your colleague agree to the corner direction? Yes.~~
- [ ] Which three corners will the demo use?
- [ ] Where can we get the current H-1011?
- [ ] Where can we read the exact 2025 NYS code and ADA sections for the rule files?
- [ ] Which public source can provide sidewalk width and legal grade?
- [ ] What units do the ramp dataset's measurements use?
- [ ] Which public bid prices can replace sample rates?
- [ ] Which civil engineer can review the rules and precedence decisions?
- [ ] Does the $40 API budget still apply, and what is the demo date?
- [ ] What domain and email-sending service will the app use?

Sources

Rule sources were checked through linked pages, but a person should confirm each one against the official text before it enters the rule set. Dataset figures come from the project's data report.

- NYC DDC: Junior Engineer (Design) posting
- NYC Mayor's Office for People with Disabilities: Accessible NYC 2025, built environment section
- DDC letter on non-standard ramps (search listing)
- NYC Administrative Code §19-112
- 34 RCNY §2-09
- NYC Street Design Manual: pedestrian ramps
- NYC DOT filing guidelines
- NYC DOT H-1011 presentation (2021)

- NYC Building Code §1104 (confirm edition)
- NYS Department of State: Uniform Code
- NYS Legal Memorandum LG07
- NYSDOT Highway Design Manual Revision 104
- U.S. Access Board: DOT's PROWAG adoption
- Congressional letter to DOJ on PROWAG (Feb 2026)
- Open Library Developer API overview
- AACE class ranges, Lawrence Berkeley Lab
- OSHA small entity silica guide
- NYC DOB: Local Law 196 summary